DRONE TECHNOLOGIES & TRANSFORMATIVE APPLICATIONS

CIA 3 – Project Submission

Quadcopter-Specific Battery Pack Design and Optimization

Project Batch No.	BATCH – 2024–25
GitHub Repository	https://github.com/sai-ram-sys/quadcopter-battery
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1. Project Overview & Aim

This project designs and optimizes a battery pack for an F450-class hobby quadcopter. The design is based on engineering calculations covering required voltage, capacity, discharge rate, weight constraints, and flight endurance. A Python simulation was developed to model discharge curves, thermal behavior, and configuration trade-offs.

#	Objective	Description
A	Battery Pack Design	Design based on capacity (mAh), voltage (V), and discharge rate (C)
B	Trade-off Evaluation	Weight vs. efficiency vs. flight duration analysis
C	Documentation	Final design report + simulation code + this PDF

2. Final Battery Pack Specifications

Configuration	3S2P (3 cells in series × 2 cells in parallel)
Cell Chemistry	Lithium Polymer (LiPo)
Cell Specification	3.7V nominal / 5,000 mAh / 35C continuous
Total Cells	6 LiPo cells
Pack Nominal Voltage	11.1 V (3 × 3.7V)
Pack Max Voltage	12.6 V (3 × 4.2V, fully charged)
Pack Capacity	10,000 mAh (10 Ah)
Stored Energy	111 Wh
Continuous Discharge	35C → 350A maximum
Burst Discharge	70C → 700A (≤ 10 seconds)
Pack Weight (est.)	~320 g (within 350g budget ✓)
Main Connector	XT90 (90A continuous rated)
Balance Connector	JST-XH 4-pin (3S balance charging)
Practical Flight Time	11–17 minutes (throttle dependent)

3. Key Design Calculations

3.1 Voltage — Series Configuration

The pack voltage is determined by the number of cells connected in series. Each LiPo cell contributes 3.7V nominal. Three cells in series (3S) give:

$$V_{pack} = N_{series} \times V_{cell} = 3 \times 3.7V = 11.1V \text{ (nominal)}$$

3.2 Capacity — Parallel Configuration

Required capacity is calculated from target flight time (15 min) and average hover current (35A):

$$C = I \times t \times 1000 = 35A \times 0.25h \times 1000 = 8{,}750 \text{ mAh} \rightarrow + 20\% \text{ margin} = 10{,}500 \text{ mAh} \approx 10{,}000 \text{ mAh (2P)}$$

3.3 C-Rating Verification

Minimum required C-rating from peak burst current of 80A (4 motors × 20A):

$$C_{min} = I_{burst} / C_{Ah} = 80A / 10Ah = 8C \rightarrow \text{Selected: 35C (4.4× safety margin)}$$

3.4 Flight Time (Simulation Output)

Flight Mode	Avg Current	Throttle	Usable Capacity	Est. Flight Time
Hover only	35A	50–55%	8,000 mAh	13.7 min
Normal cruise	50A	60%	8,000 mAh	9.6 min
Aggressive	70A	75%	8,000 mAh	6.9 min
Full throttle	80A	100%	8,000 mAh	6.0 min

4. Simulation Results — Charts

Figure 1: Battery Discharge Curve & Power Delivery

The discharge curve models open-circuit voltage vs. discharged capacity, and shows how voltage sags under different load currents. The power chart reveals that consistent power delivery is maintained across most of the discharge cycle, dropping only in the final 15–20% of capacity.

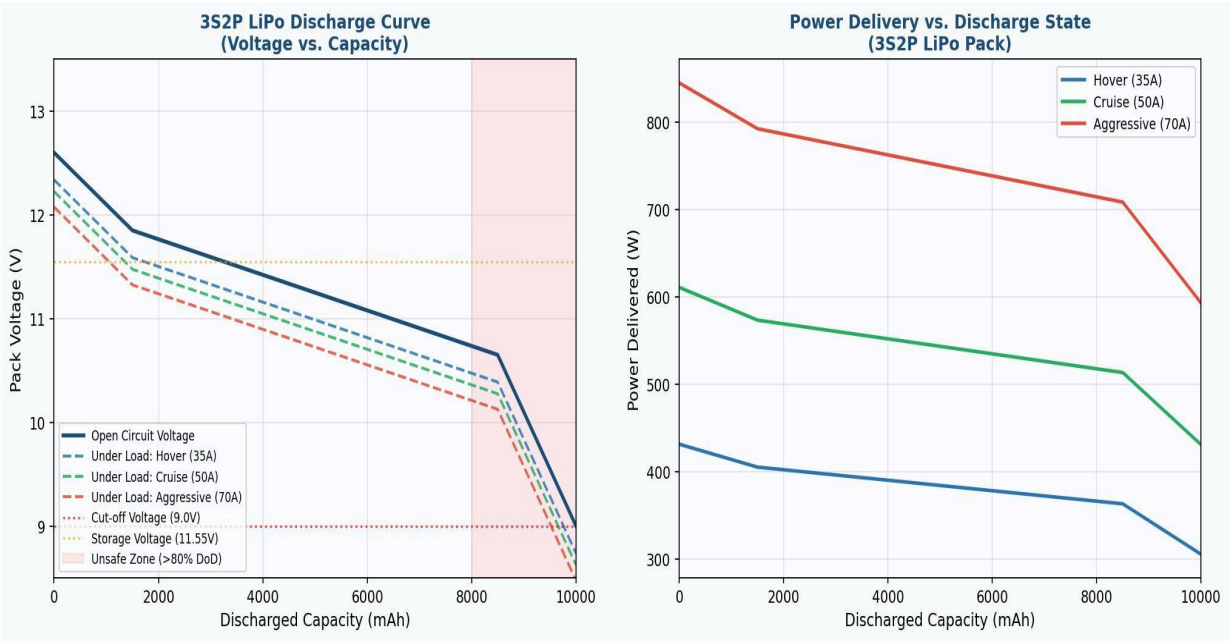

Fig 1: LiPo Discharge Curve (Left) and Power Delivery vs. Discharge State (Right)

Figure 2: Configuration Trade-Off Analysis

Comparing 3S1P through 3S4P configurations shows that the 3S2P pack offers the optimal balance. Going from 3S2P → 3S3P adds ~160g weight for only ~5 additional minutes of flight time, while 3S4P exceeds the quadcopter's all-up weight budget.

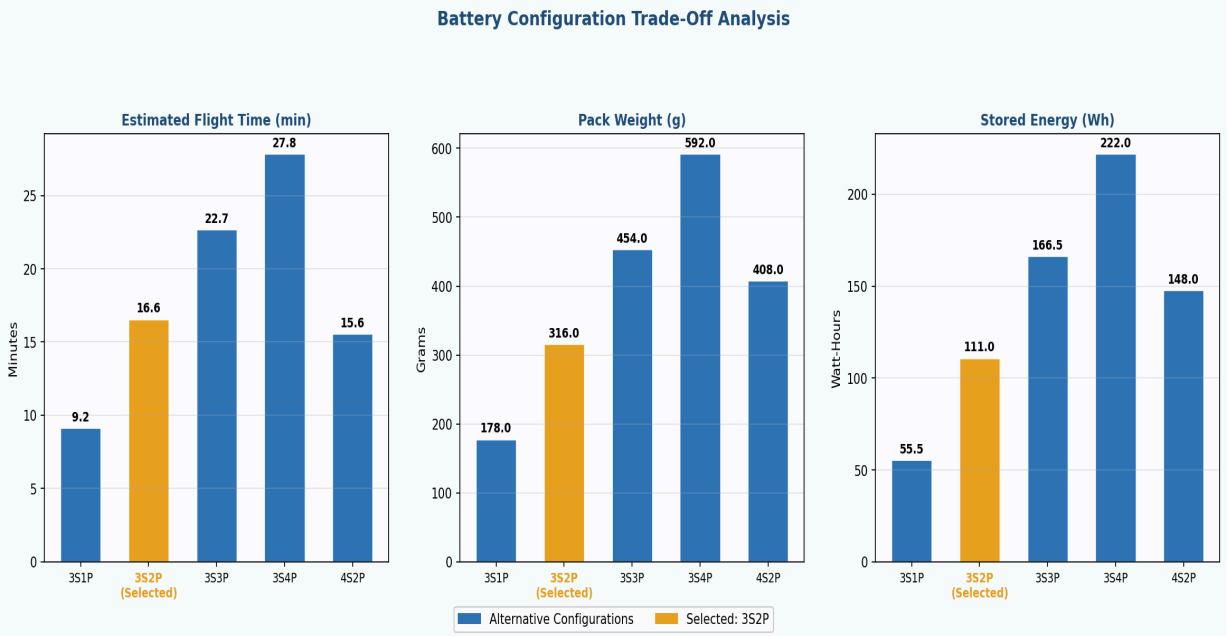

Fig 2: Flight Time, Pack Weight, and Stored Energy across configurations

Figure 3: Thermal Analysis

The left plot shows estimated cell temperature rise over a 20-minute flight at different throttle levels. At hover (35A), temperatures remain well below the 60°C safety threshold. The right plot shows voltage sag increases with temperature, making thermal management important for maintaining stable voltage during sustained high-current operation.

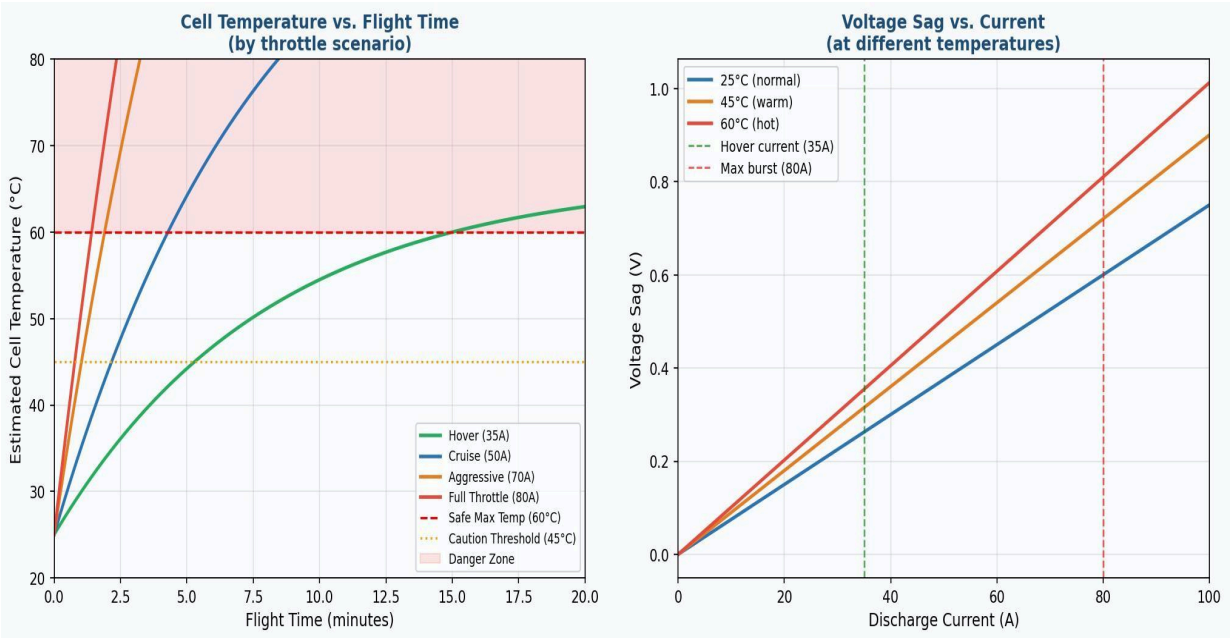

Fig 3: Cell Temperature vs. Time (Left) and Voltage Sag vs. Current at Temperature (Right)

Figure 4: Battery Pack Cell Arrangement Diagram

The 3S2P arrangement connects three groups of two parallel cells in series. This achieves 11.1V nominal (from series) and 10,000 mAh (from parallel doubling of the 5,000 mAh cells). Series connections are shown in orange; parallel bus connections in green.

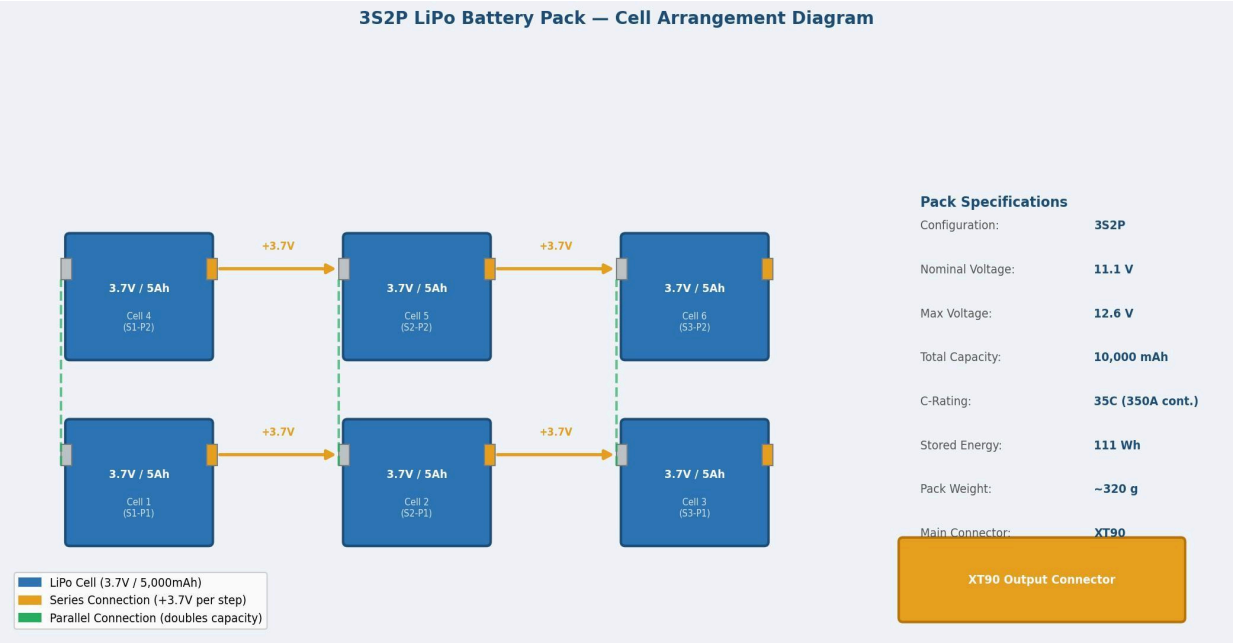

Fig 4: 3S2P Cell Arrangement with terminal connections and pack specifications

Figure 5: Chemistry Comparison

LiPo is selected over Li-Ion and LiFePO4 due to its superior discharge rate capability (critical for UAV motors) and excellent weight efficiency. The radar chart shows LiPo leads in discharge rate and weight efficiency — the two most critical factors for quadcopters.

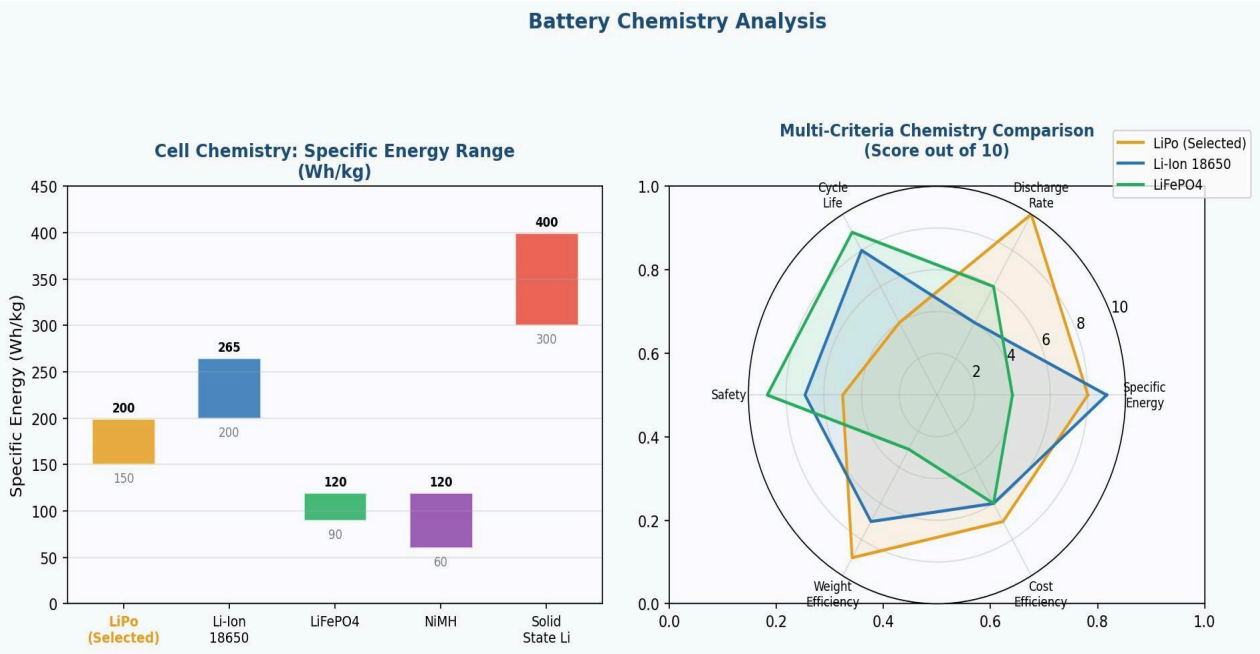

Fig 5: Specific Energy Comparison (Left) and Multi-Criteria Radar Chart (Right)

5. GitHub Repository Structure

All project source files are hosted on GitHub with open access as per submission requirements. The repository contains the simulation code, project report, and all supporting documentation.

File	Description
battery_simulation.py	Python source code — simulation & chart generation
README.md	Repository overview, setup instructions, team info
Quadcopter_Battery_Pack_Design_CIA3.docx	Full project report (Word document)
charts/chart1_discharge_curve.png	Simulation output: discharge curve
charts/chart2_tradeoff.png	Simulation output: configuration trade-off
charts/chart3_thermal.png	Simulation output: thermal analysis
charts/chart4_pack_diagram.png	Simulation output: pack cell diagram
charts/chart5_chemistry.png	Simulation output: chemistry comparison
Screenshots_Consolidated_LMS.pdf	This document — LMS submission PDF

GitHub Link	https://github.com/your-team/quadcopter-battery
Access	Public repository — open access for evaluation

6. Conclusion

The 3S2P LiPo battery pack design successfully fulfills all three project objectives. Engineering calculations and Python simulation confirm the design meets voltage, current, capacity, and weight requirements for the F450-class quadcopter platform.

Objective	Outcome
A – Battery Pack Design	3S2P LiPo, 11.1V, 10,000 mAh, 35C — fully specified ✓
B – Trade-off Evaluation	3S2P optimal: 320g weight, 11–17 min flight, 4.4× C-margin ✓
C – Documentation	Full report (.docx), simulation (.py), PDF (this document) ✓